SENTINEL · DEFENDER XDR · AWS

Cloud Incident Response & SOAR Automation Strategy



Standard Operating Procedures (SOPs) & Logic App Noise-Reduction Frameworks

TIER-2 / SHIFT LEAD OPERATIONAL VIEW

Dual-Playbook Model

Confluence SOPs + Logic App Automation

~35%

Queue Noise Reduction via Automated Baseline Suppression

Forensic Integrity

EDR Network Isolation over Destructive VM Shutdowns

GOVERNANCE MODEL

Operational Framework: Threat Containment vs. Asset Governance

Active Threat Containment

SOC OWNED

SCOPE

Active C2, execution, credential harvesting, external data exfiltration.

ACTION

Immediate SOC ownership; execute EDR host isolation and credential revocation.

FORENSIC MANDATE

Sever network sockets via EDR; NEVER power off or terminate VMs (preserves volatile RAM and running process trees).

Hygiene & Asset Governance

SOAR AUTOMATED

SCOPE

Untagged resources, missing backup policies, unencrypted storage buckets.

ACTION

Bypasses SOC queue entirely via automated Logic App workflows.

AUTOMATED WORKFLOW

Traces creation principal via audit logs, tags asset dynamically, and routes a Jira/ServiceNow ticket directly to the engineering owner with SLA enforcement.

MODULE 01 · OVERVIEW

Module 1: Cross-Cloud Identity Compromise & Storage Exfiltration

STEP 1

ENTRY VECTOR

Entra ID MFA fatigue / token theft

→ Session hijacking

STEP 2

CROSS-CLOUD PIVOT

Federated SSO traversal to AWS Console

→ Long-term IAM Access Key creation

`CreateAccessKey`

STEP 3

TARGET EXFILTRATION

Outbound data exfiltration from cloud storage

`GetObject`	AWS S3
`GetBlob`	Azure Storage

BUSINESS RISK

Silent exfiltration across multi-cloud storage from unmanaged external IPs.

MODULE 01 · DUAL PLAYBOOK

Module 1 Playbook: Confluence SOP & Logic App SOAR

Confluence / Manual Analyst SOP

Documented triage and scoping

1. **Triage Entry Vector:** Query `SigninLogs` for `RiskState == "atRisk"` and inspect `LocationDetails` and `UserAgent`.
2. **Cross-Cloud Scoping:** Query `AWSCloudTrail` for `EventName` in (`"CreateAccessKey"`, `"AssumeRoleWithSAML"`) linked to user UPN.
3. **Exfiltration Audit:** Query `StorageBlobLogs` (`CallerIpAddress`, `OperationName == "GetBlob"`) or `AWSCloudTrail` (`GetObject`) to calculate total outbound bytes.
4. **Manual Escalation:** If exfiltration exceeds 1GB, notify Incident Commander and initiate forensic review.

Logic App SOAR Automation

Noise reduction and automated response

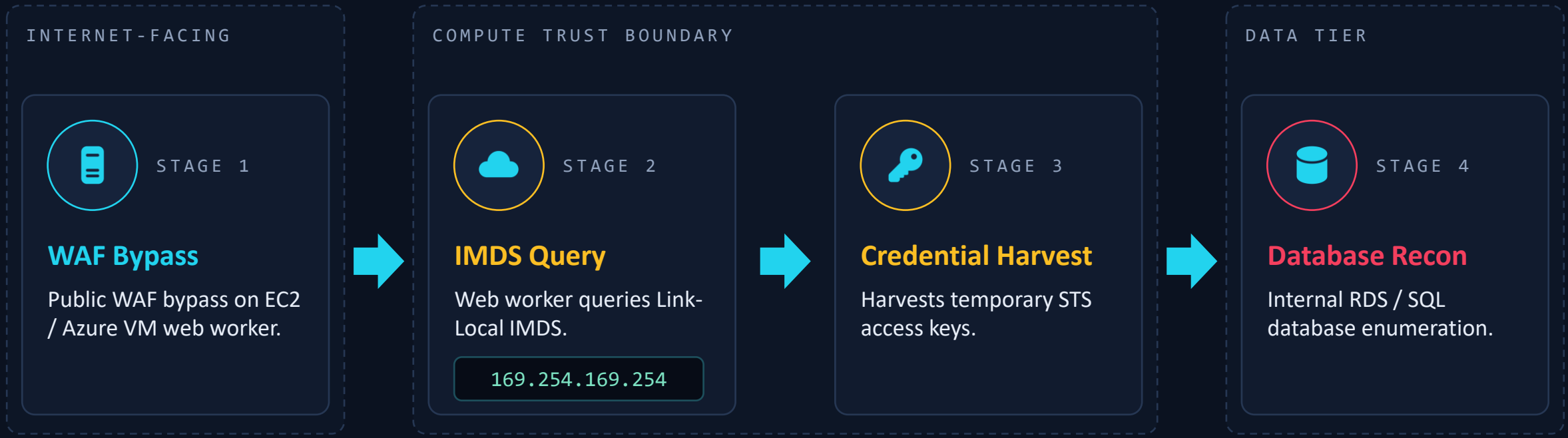
1. **Trigger:** Sentinel Alert (High-Risk Identity / Multi-Cloud Anomaly).
2. **Noise Reduction:** Check if source IP belongs to corporate VPN/Proxy ranges. Auto-close as Benign FP if matched (~35% queue cut).

   `MATCH → AUTO-CLOSE FP` `EXTERNAL IP → STEP 3`

3. **Auto-Remediation:** If external IP, revoke Entra ID refresh tokens (`Revoke-AzureADUserAllRefreshToken`), disable AWS IAM Access Keys via AWS SDK, and post a structured incident card to the Teams SOC channel.

MODULE 02 · OVERVIEW

Module 2: Compute RCE, IMDS Theft & DB Reconnaissance



FORENSIC INTEGRITY RULE: Sever network sockets via EDR host isolation. **NEVER power off or terminate the VM**—powering down destroys volatile RAM, process trees, and injected memory artifacts.

MODULE 02 · DUAL PLAYBOOK

Module 2 Playbook: Confluence SOP & Logic App SOAR

Confluence / Manual Analyst SOP

Documented triage and scoping

1. **Process Execution Triage:** Inspect EDR logs (`DeviceProcessEvents`) for `InitiatingProcessFileName` in (`nginx`, `w3wp.exe`, `www-data`) spawning shell binaries or querying `169.254.169.254`.

2. **Telemetry Pivot:** Query `AWSCloudTrail` for `UserIdentityType == "AssumedRole"` API calls from external, non-AWS IP addresses.

3. **Network Scoping:** Check `VPCFlowLogs` / `AzureDiagnostics` for outbound connection attempts to internal database subnets.

Logic App SOAR Automation

Noise reduction and automated response

1. **Noise Suppression:** Auto-suppress alerts matching scheduled vulnerability scanner subnets or DevOps maintenance windows.

   MATCH → AUTO-SUPPRESS

   NO MATCH → CONTAIN

2. **Auto-Containment:** Execute CrowdStrike/Defender EDR network isolation, force IMDSv2 enforcement across instance, revoke STS assumed role sessions, and trigger automated EBS/OS disk snapshotting for forensics.

   EDR NETWORK ISOLATE

   ENFORCE IMDSv2

   REVOKE STS SESSIONS

   SNAPSHOT EBS / OS DISK

MODULE 03 · OVERVIEW

Module 3: Unmanaged Asset Governance & DB Snapshot Hijacking

ATTACK CONTEXT

Adversary creates RDS snapshot (`CreateDBSnapshot`) and alters permissions (`ModifyDBSnapshotAttribute` / `SharedSnapshot`) to exfiltrate database copies to an external AWS account.

STRATEGIC GOVERNANCE MODEL

Asset Hygiene

SOAR AUTOMATED

ALERT TYPE

Untagged / unmanaged resource alerts

HANDLING

Handled via automated SOAR ticketing to engineering.

Active Threat

SOC OWNED

ALERT TYPE

Cross-account DB snapshot exfiltration

HANDLING

Handled via SOC incident command and automated session revocation.

MODULE 03 · DUAL PLAYBOOK

Module 3 Playbook: Confluence SOP & Logic App SOAR

Confluence / Manual Analyst SOP

Documented triage and scoping

1. Query `AWSCloudTrail` filtering for `EventName` in (`CreateDBSnapshot`, `ModifyDBSnapshotAttribute`, `SharedSnapshot`).
2. Parse `RequestParameters` to extract target `attributeValue` (External AWS Account ID).
3. Verify whether the target account ID exists on the approved Enterprise Whitelist.
4. **Manual Escalation:** Contact cloud infrastructure team to verify if cross-account sharing was an authorized migration.

Logic App SOAR Automation

Noise reduction and automated response

A **Path A:** Hygiene Alert on Untagged Asset

- → Trace creating principal via `AWSCloudTrail` / `AzureActivity`
- → Dynamically tag asset
- → Route Jira/ServiceNow ticket directly to engineering owner with SLA

B **Path B:** Active Snapshot Exfiltration Threat

- → Revoke active IAM session keys immediately
- → Execute `ModifyDBSnapshotAttribute` API call to revert external sharing permissions
- → Alert Incident Commander via Teams / PagerDuty

KQL REPOSITORY

Production-Ready Sentinel KQL Detection Queries

SNIPPET 01

Identity & Exfiltration

SigninLogs · StorageBlobLogs

```
SigninLogs
| where RiskState == "atRisk"

StorageBlobLogs
| where OperationName in ("GetBlob", "GetBlobProperties")
| extend SourceIP = tostring(split(CallerIpAddress, ":")[0])
| summarize TotalBytes = sum(ResponseBodySize) by SourceIP
| where TotalBytes > 1000000000
```

SNIPPET 02

IMDS Credential Theft

DeviceProcessEvents

```
DeviceProcessEvents
| where InitiatingProcessFileName in~ ("nginx.exe", "w3wp.exe", "www-data")
| where ProcessCommandLine contains "169.254.169.254"
```

SNIPPET 03

Database Snapshot Sharing

AWSCloudTrail

```
AWSCloudTrail
| where EventName in ("CreateDBSnapshot", "ModifyDBSnapshotAttribute")
| extend RequestParams = parse_json(RequestParameters)
| extend TargetAccount = tostring(RequestParams.attributeValue)
```

EXECUTIVE SUMMARY

SOC Operational Summary & Metrics Impact

~35%

Queue Noise Reduction

Achieved by auto-suppressing verified corporate proxy IPs, vulnerability scanners, and automated hygiene ticket routing.

100%

Volatile RAM Retention

Enforced via EDR socket isolation policies over destructive VM power-offs, protecting process memory artifacts.

Sub-5 min

Response SLA

Driven by Logic App automation for Entra ID token revocation, IAM key deactivation, and RDS snapshot permission rollbacks.

Dual-playbook model: Confluence SOPs drive documented triage and scoping; Logic App playbooks drive noise reduction and automated response.